CSC311: Introduction to Machine Learning

Project Report

Optional Creative Project Title

Submitted by

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1 Executive Summary

We propose to classify a sport as soccer, baseball, or basketball using survey responses about its accessibility, playing characteristics, and cultural associations. The dataset contains 870 responses from 290 respondents, with 290 examples of each sport. We will compare regularized softmax regression, random forests, and Naive Bayes to investigate how linear relationships, nonlinear interactions, and textual evidence contribute to classification. We hypothesize that softmax regression will provide a strong balance between predictive performance and interpretability when trained on encoded survey responses and TF-IDF text features. We will select the final model using validation results grouped by respondent and assess it using accuracy and macro-F1. A projected test accuracy will be reported after completing these experiments.

2 Data Exploration

The task is supervised classification with three balanced classes: soccer, baseball, and basketball. Each of the 290 respondents contributes three observations. The dataset contains 12 survey features, a respondent identifier, and a sport label. The features include a numerical accessibility rating, three ordered agreement questions, three multi-select questions, four free-text association questions, and one free-text willingness-to-pay question.

The respondent identifier will be used to construct data splits but excluded from the predictive features. All observations from the same respondent will remain in the same split to prevent the model from benefiting from respondent-specific writing styles or response patterns during evaluation.

Feature preparation TABLE HERE

Numerical features will be standardized for softmax regression. Text questions will be vectorized separately so that the same word appearing in different questions can receive different weights. Missing values and unusual responses An initial check found between 7 and 17 empty responses per feature. The song question has the most empty responses: 17 out of 870, approximately 2.0%. These counts only include blank entries; further exploration will check for placeholders such as “N/A” and other non-informative responses. We will inspect numerical values for invalid entries and extreme values. For willingness to pay, we will examine whether a $\log(1 + \text{price})$ transformation improves the representation of large positive values. Missingness indicators will allow models to use whether a response is absent as a potential feature. Planned exploration After establishing the data splits, we will examine feature distributions and relationships with the label using only the training partition. We will compare numerical and agreement-response distributions across sports, plot multi-select frequencies by class, and examine class-associated words in the text responses. We hypothesize that songs, colours, foods, playing locations, and seasons may help distinguish the sports. These are proposed relationships to investigate, rather than findings established by the initial inspection.

3 Methodology

Data splitting and validation We will reserve approximately 20Imputation values, scaling parameters, categorical vocabularies, and text vocabularies will be learned separately from each fold's training data. The internal test set will remain unused during feature selection and hyperparameter tuning. Model 1: Regularized softmax regression Softmax regression will predict the probabilities of the three sports from a combination of structured features and TF-IDF text features. It is appropriate as a relatively simple model for a dataset with many potentially sparse text features. L2 regularization will help control overfitting. We will tune the regularization parameter, compare unigrams with unigrams plus bigrams, and vary the minimum document frequency used to retain text features. We will inspect model coefficients to identify which responses contribute most strongly to each prediction. Model 2: Random forest A random forest will model nonlinear relationships and interactions between responses. For example, the relationship between a season response and a sport may depend on the locations selected by the respondent. We will first evaluate the forest using structured features, then test whether adding a restricted text vocabulary improves validation performance. We will tune maximum tree depth, minimum samples per leaf, and the number of features considered at each split. Candidate settings could include 100 and 300 trees, maximum depths of 5, 10, and unrestricted depth, and minimum leaf sizes of 1, 3, and 5. Model 3: Naive Bayes Multinomial Naive Bayes will provide a probabilistic text baseline using nonnegative word-count or TF-IDF features from the free-text questions. It is computationally inexpensive and can combine many weak word-level signals. We will tune the smoothing parameter over values such as 0.1, 0.5, 1, and 2, and compare word counts with TF-IDF. Its conditional-independence assumption may limit performance when words or survey responses are strongly related. Fair comparison and feature experiments All model families will use the same respondent-based folds and receive hyperparameter tuning before comparison. We will compare structured-only, text-only, and combined representations where appropriate to determine how much each feature group contributes. The supplied kNN implementation can serve as an additional reference baseline. We will also report a constant-class baseline, which achieves 33.3% accuracy on this balanced dataset. Evaluation We will report accuracy, macro-F1, per-class precision and recall, and confusion matrices. Accuracy measures overall correctness, while macro-F1 gives each sport equal importance and helps reveal uneven performance across classes. We will summarize the mean and standard deviation of cross-validation scores. Our primary selection metric will be mean validation accuracy, with macro-F1 and consistency across folds informing close comparisons.

4 Results

For the proposal, use planned analysis rather than invented scores:

We will summarize the tuned models' cross-validation performance in the following table:

TABLE

We will analyze confusion matrices to determine which sports are most frequently confused. We will also examine whether errors are associated with missing responses, uncommon song titles, ambiguous text, or similar structured answers.

We hypothesize that softmax regression will perform well because regularization can help it handle sparse text features. Random forests may perform better if interactions among structured responses are especially informative. Naive Bayes will establish how effectively word-level evidence alone distinguishes the classes.

After selecting the model and preprocessing configuration, we will evaluate the selected pipeline once on the internal test set. We will use this result, supported by cross-validation stability, to justify a single projected accuracy for the unseen course test set. No defensible numerical estimate is available yet.

5 Contributions and Learning

This section should contain a list of 3–4 bullets, one for each student. Each student should write at most three sentences describing their main contributions and the most important lesson learned.

6 References

Use references whenever appropriate (e.g., when you consult textbooks, research papers, online tutorials, or external code). Cite in the text using `\cite{}`. For example, let's cite the two references provided in the `.bib` file [**?**, **?**].

appendices.

A Guiding Questions for Data Exploration

These questions support Section 2. Work through them in order, and record your answers for our dataset in that section rather than here. This appendix is part of the template instructions, so comment it out before you submit, as you do with the rest of the instructions. what is the number of classes?

1. Determine Input to Model
   (a) For each example, what features do we have? What are the types of the features (numerical, categorical, or text)?

   (b) In the released data set, each row has a unique ID identifying which student provided the data. Why should we not use this unique ID as a feature? highly skewed? Is there enough variation? Are there outliers?

2. Data Splitting
   (a) Why is it a good idea to split the provided data into a training set and a test set?

   (b) Why is it a good idea to do the training and test set splitting based on the unique ID identifying each respondent?

   (c) What is the role of the validation set and why do we need this third split?

   (d) Why should we only perform data exploration on the training set and not on the validation or the test sets?

3. Feature Distributions One reason for exploring the distributions of features is to get to know the data before building any model. This exploration can help us catch common issues such as outliers, skewed distributions, class imbalance, missing values, and large difference in feature scales.
   (a) What should we do when exploring the distributions of the features?

   (b) Suppose a feature takes the same value for almost every respondent. Why is this feature worth finding, and what would you do about it?

   (c) Suppose two features are highly correlated, such as two Likert questions that respondents tend to answer the same way. How would you test whether two features are highly correlated, and why does a highly correlated pair cause trouble?

4. Class Imbalance
   (a) What is class imbalance? Why is class imbalance problematic?

   (b) Does the dataset have a class imbalance issue? If so, how severe?

   (c) What are some common strategies to address class imbalance?

5. Missing Values
   (a) First, determine if the dataset contains missing values. Report the percentage of values missing for each feature in a table. A missing value can take many forms such as a null value, an empty string, an empty list, “n/a”, or “-”. Normalize all of these to a single missing marker before you count, or you will underestimate how much is missing.

   (b) Next, for each feature with a substantial missing rate, try to determine why the values are likely missing. Different explanations may require different fixes. What are some possible explanations for missing values?

   (c) Since you never see the missing value, it may not be possible to verify which explanation holds. However, it would be useful to determine whether the missingness of a feature is related to the label or not. If the missingness is related to the label, a missing value itself provides a predictive signal. What is a common strategy for determining this?

   (d) What are some common strategies to handle missing values? One is called imputation.

   (e) When is it appropriate to use each strategy above to handle missing values?

   (f) Suppose that you decide to impute the median value for a numeric feature. Why is it problematic to do the imputation on the validation and test sets?

   (g) Finally, decide how missing values should be handled for each feature in our dataset, and record your decision.

6. Outliers
   (a) What observation might cause you to suspect that certain values are outliers?

   (b) How do I determine whether a feature value should be considered an outlier?

   (c) What are some strategies to handle outliers?

7. Feature Engineering
   (a) How can we convert the responses to a multi-select question into a feature?

   (b) The survey has a few Likert-scale questions that ask respondents to indicate their level of agreement to a statement using an ordered set of options (e.g., from Strongly Disagree to Strongly Agree). We can convert the Likert-scale responses to features by modeling them as ordinal values or one-hot encoded categories. What are some advantages and disadvantages of each approach?

   (c) At least one free-text question asks for a number. How would you turn such a text response into a number, and when would you declare a response unparseable?

   (d) What are some strategies to convert the free-text responses to numeric features?

(e) One common technique for processing the free-text responses is TF-IDF vectorization. Explain this technique in detail. Why is this technique suitable for processing the free-text data?

(f) How should we handle a word that appears at prediction time but never in the training split?

(g) After encoding all the features, you recorded that there are 1100 features and 300 examples. How would the ratio between these two inform your model choices? Why would you choose a simpler model with strong regularization over a larger model?

8. Feature Scaling

(a) Two common choices for feature scaling are standardization and min-max scaling. Explain each in one sentences. Why do we need to perform feature scaling for our dataset, which has Likert-scale, multi-hot encoded, and word count features?

(b) Is each model below affected by feature scaling? Briefly explain.

- K-nearest neighbours
- Decision trees and random forest
- Softmax regression trained using gradient descent
- L2 Regularization
- Naive Bayes

(c) You decide to standardize a feature. Which split do you use to compute the mean and standard deviation, and why?

9. Which Features Are Predictive Next, we would like to determine whether a feature is useful by looking at the feature together with the label. This analysis could also tell you which model families are worth trying and help you justify your model choices in the Methodology section.

(a) How do you examine the relationship between a Likert-scale or numeric feature and the label?

(b) How do you examine the relationship between a multi-select or free-text feature and the label?

(c) Why should you only do this analysis on the training split and not on the validation or test splits?

(d) How can you delete whether a feature is worth keeping by comparing the difference in mean feature values between classes with the spread of feature values within a class? Consider a Likert-scale feature on a scale of 1 to 5. Is the feature worth keeping if two classes have means 3.5 and 3.8 while the standard deviation of answers within a class is 1.2? What about if two classes have means 2.1 and 4.3 while the standard deviation of answers within a class is 0.6?

(e) What does the shape of the relationship tell you about which model families to try? Consider the three scenarios below. Discuss the model families you would try for each one.
   i. The classes separate well along one feature at a threshold.
   ii. The middle values of a Likert feature belong to one class while both extremes belong to another.
   iii. After vectorizing the free-text responses, you have several hundred word-count features. No single word separates the classes, but each word in a response carries a small amount of evidence.

(f) Finally, write down the features you expect to be the most predictive, and name the plot or the number that led you to each one.

B Guiding Questions for Methodology

These questions support Section 3. Work through them in order, and record your answers for our dataset in that section rather than here. This appendix is part of the template instructions, so comment it out before you submit, as you do with the rest of the instructions.

1. Choosing Model Families
   (a) Based on the size of the data set, should we consider fine-tuning a large language model or starting with simpler models? What are some advantages and disadvantages of each approach?

   (b) Which findings from your data exploration would support choosing softmax regression? Which findings would support choosing a decision tree instead?

2. Training Details
   (a) Suppose you train a multilayer perceptron. Which choices would another student need to know to reproduce your results exactly?

   (b) Your softmax regression model reaches 95% accuracy on the training set but only 70% on the validation set. What does this gap tell you, and which regularization or early stopping strategies could you try?

3. Validation
   (a) What is cross validation? If we want to use cross validation, why should we use grouped cross validation, grouped by the unique ID?

   (b) Pre-processing steps such as normalization and text vectorization learn values from the data, like a mean or a vocabulary. Which part of the data should these values be computed from during cross validation, and why?

4. Hyperparameter Tuning
   (a) Which hyperparameters would you tune for a decision tree, and how would you choose the range of values to try?

(b) You try maximum depths from 2 to 10 for a decision tree, and the validation accuracy is highest at a depth of 10. Is your search finished? What would you do next?

(c) After choosing your final model and its hyperparameters, should you retrain it on the training and validation sets combined? What do you gain, and what do you lose?

5. Comparing Model Families Fairly
   (a) With default hyperparameters, softmax regression reaches 78% validation accuracy and a decision tree reaches 72%. You tune softmax regression up to 82% and choose it as your final model. What is wrong with this process, and how would you fix it?

   (b) Across five folds, model A has a mean validation accuracy 1% higher than model B, but the accuracy of each model varies by about 3% from fold to fold. Can you conclude that model A is better? What else could help you decide?

6. Evaluation Metrics
   (a) Suppose one class makes up 60% of the training set. What accuracy does a model that always predicts this class achieve? Which metric would reveal that this model is useless?

   (b) How do you compute precision, recall, and F1 for a 3-class problem? What is the difference between macro-averaging and weighted averaging, and when would you prefer each one?

7. Additional Techniques
   (a) Suppose you apply PCA to several hundred word-count features. How would you decide how many components to keep, and which part of the data should you fit PCA on?

   (b) How could you use a clustering method such as k-means to explore the free- text responses? How would you tell whether the clusters improve your predictions?